Data Modelling:

Clustering

Machine learns via two main styles

II. UNSupervised Learning:

- Process of an algorithm to model the underlying structure or distribution in the data in order to uncover previously unknown patterns in data
- To bring similar observations together
- There is NO teacher-student relationship
- There is NO feeding of **labels/targets** during training.
- NO way to determine how accurate is the outcome (no Ground Truth to verify)

We are different.
But you need to
study us in detail
to know why

Supervised Vs Unsupervised Learning

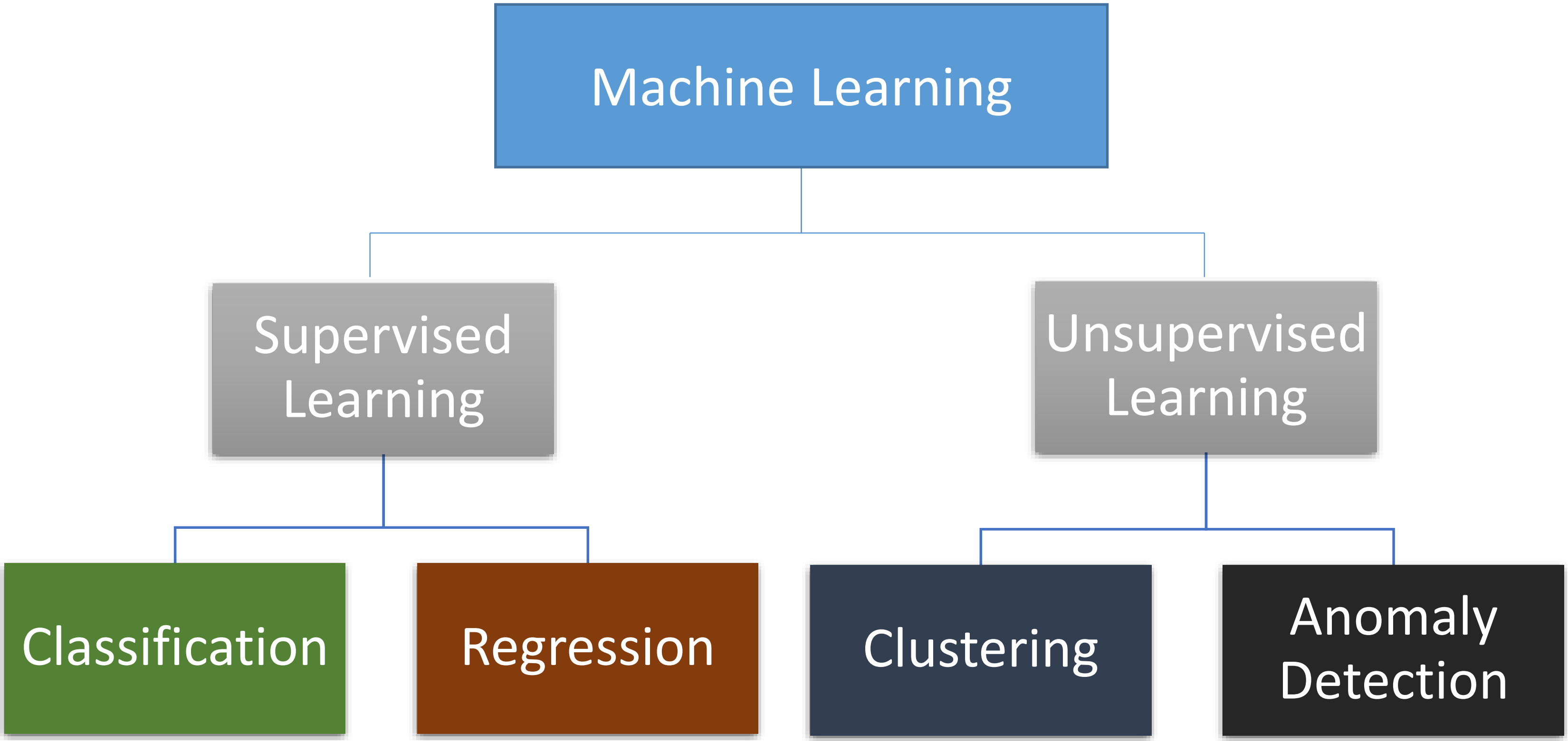
	Supervised Learning	Unsupervised Learning
Advantage	Label acts as the ground truth to validate model performance	**Unlabelled data are easily available**
Disadvantage	Difficult/expensive to obtain labelled data	**More subjective** than supervised learning, as there can be no simple goal. Won't be able to know if the model is right or wrong, e.g. users may interact more frequently with some particular product listings, BUT that does not mean the user 'like' those products-'like' or 'dislike' is a label.

Supervised Vs Unsupervised Learning

- The ability to verify if the model is accurate or not, makes supervised learning applicable to most practical needs in the industry.

- However there are also some applications that Unsupervised Learning can perform better.

Different families of algorithms

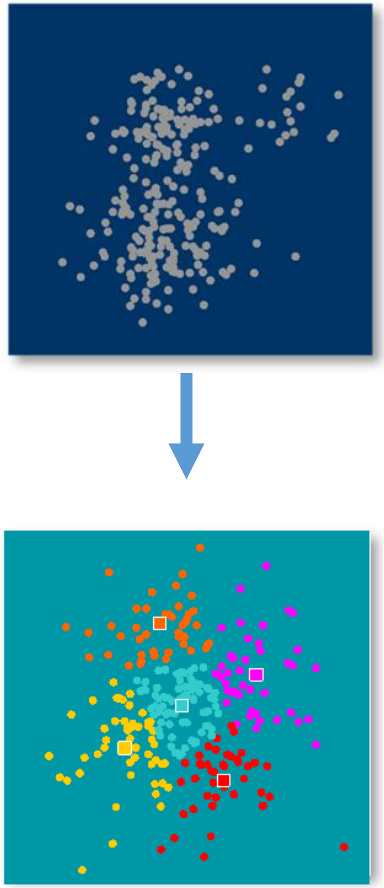


k-means Clustering

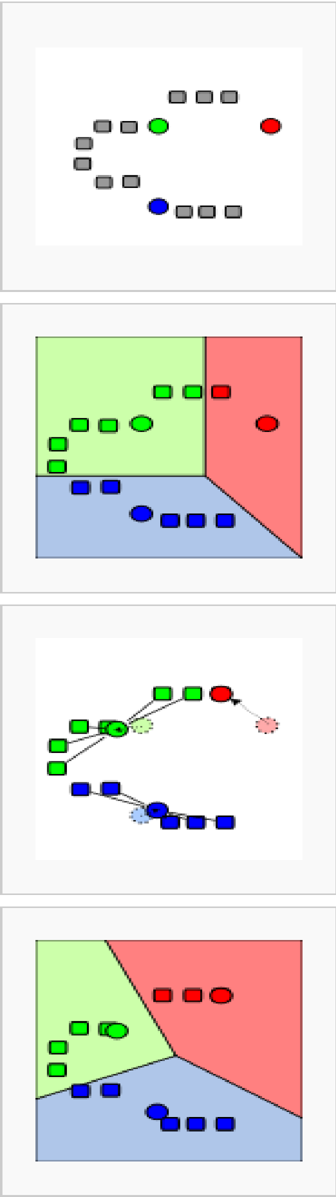
- One of the more commonly used methods for clustering is the **k-means algorithm.**

- Common use of this is to **observe patterns and find clusters/groups.**

- Examples:
  - Market segmentation
  - Profiling claimants
  - Fraud Detection



The Workflow



Step 1: Choose k (the number of desired clusters) random positions in the input space

Step 2: Assign them as cluster centroids

Step 3: For each data instance, compute the distance to each centroid

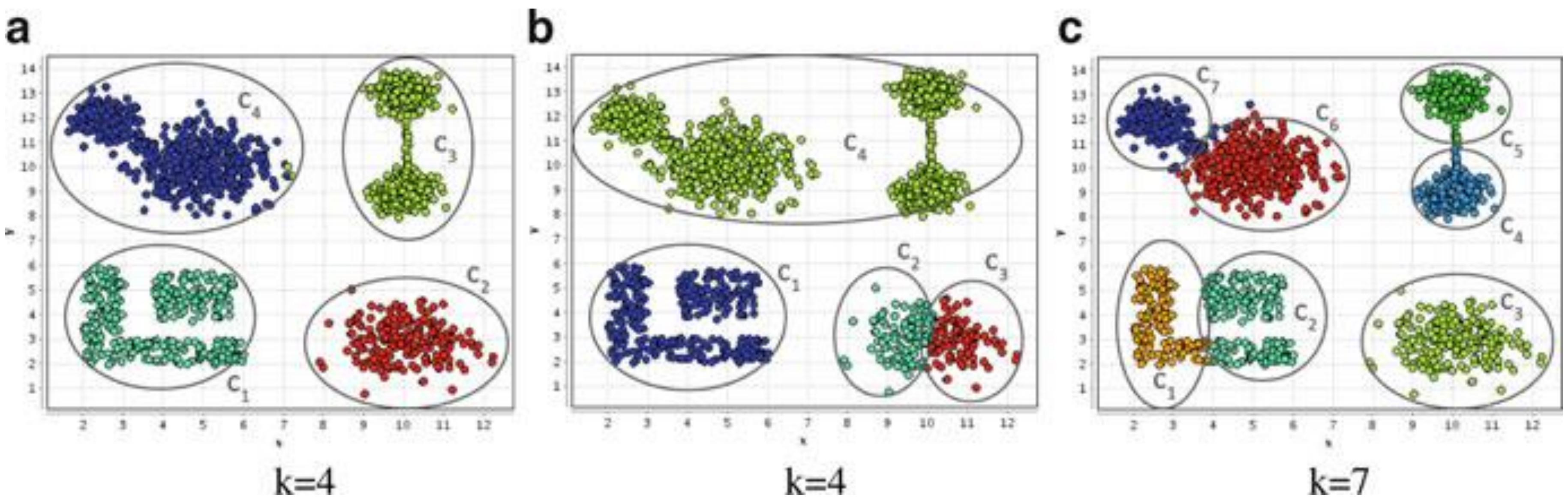
Step 4: Assign the data instance to cluster with min distance

Step 5: Compute mean of each cluster and assign new cluster centroids

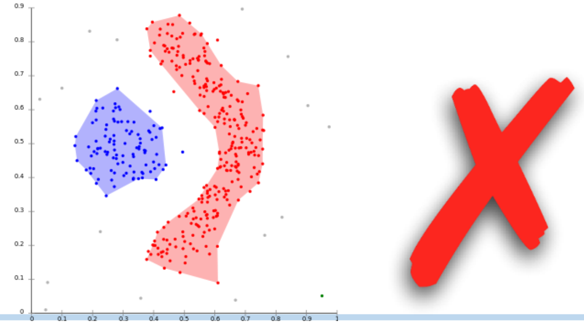
Step 6: Repeat Steps 3-5 until convergence

Challenges of k-means

- What is the suitable k (no. of clusters)?



- Does not work well for overlapping data points



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